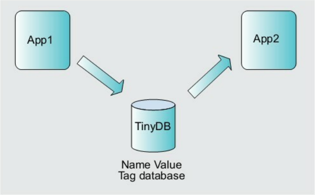

Figure 1: Apps communicate using TinyDB

Tag name	Explanation
QuestionSetList	A comma-separated list of all QuestionSets created using this app; e.g., set1, set2, set3, etc
<QuestionSet><n>Q	A text field representing the nth question in <QuestionSet>
<QuestionSet><n>A	A text field representing the answer to the nth question in <QuestionSet>
<QuestionSet>nTotalQ	A number field representing the total number of questions in <QuestionSet>

Table 2: The design of tags

The main screen

Now we come to the main screen. In each of these screens we first take a look at the elements we use from the palette in the 'design' view and what their names and values are to be. We will then present the code in the 'blocks' view and explain to an extent the role played by each code piece. In the main screen, which could not be renamed from 'Screen1' as far as I could make out, we see the elements described in Table 3 with their properties. Also note that TinyDB is a non-visible component. The screen itself should look like what is shown in Figure 2.

Name	Element type	Palette group	Property	Value
Screen1	Screen	NA	Title	Create or answer questions
CreateQ	Button	User interface	Text	Create questions
An-swerQ	Button	User interface	Text	Answer questions
ClearAll	Button	User interface	Text	ClearAll
Debug	Button	User interface	Text	Debug
TinyDB1	TinyDB	Storage		

Table 3: Screen1 elements

Ideally 'Create Questions' and 'Answer Questions' can be separate apps, as you may want to offer a different service to creators of questions and provide a different app for answering. But for this article, it would be fine to use the common app with different screens to be able to illustrate how they work using TinyDB for storage and retrieval of data. The blocks for Screen1 look like what's shown in Figure 3.

The main functionality of Screen1 is only to provide an interface to control invocation of the other screens, namely CreateQuestions, AnswerQuestions and also a DebugScreen. There is one more entry to ClearAll, which basically cleans up TinyDB of all the entries. Both the ClearAll and the DebugScreen aid in debugging during development. They are not expected to play this role in a production level app. The new procedure call, which opens another screen *screenname*, is available from the *Built-in→ Control* group of blocks. It basically takes a screen name as an argument and instantiates that particular screen. As of now, it looks like using the *Back* button is necessary to come back to the main screen and to do a clean-up in the child screen. We will come to that later in the next section.

Note: It is interesting to note that the screens cannot share any variables or any other common component. Also, even a copy-paste does not work across screens, either in design mode or in blocks mode.

The CreateQuestions screen

Now we come to the CreateQuestions screen. The elements of the screen are described in Table 4. The screen itself is shown in Figure 4.

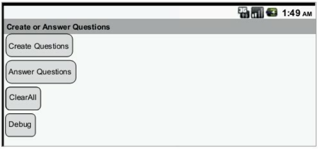

Figure 2: The main screen

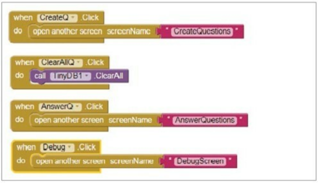

Figure 3: Blocks for main screen